

L4L 2.2. $\langle , \rangle$ v $\mathbb{R}^2$; $\langle u, w \rangle = u^T A w$

B (t.j. $b_1 \perp b_2$ vzhľadom k z.s.d.) k $\langle , \rangle$

~~musí~~ byť každé vektory kanonickú bázu $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ a $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

vech $B = \{u, v\}$ kde $u = Re_1$ a $v = Re_2$ kde R
je rotačná matica: $\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$

potom chceme $u^T A v = 0$. vech $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ (predpoklad je pos. def.)

$$\text{pot } u^T A v = (\cos \theta \quad -\sin \theta) \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \sin \theta \\ \cos \theta \end{pmatrix} \quad (a - b^2 > 0)$$

$$= a \sin \theta \cos \theta + b \cos^2 \theta - b \sin^2 \theta - c \sin \theta \cos \theta = 0$$

(stačí ukázať, že sa vždy dá nájsť taký uhol.)

$$\Leftrightarrow (a - c) \sin \theta \cos \theta + b(2 \cos^2 \theta - 1) = 0$$

$$\Leftrightarrow \frac{(a - c)}{2} \sin 2\theta + b(\cos 2\theta) = 0$$

$$\tan 2\theta = \frac{2b}{c - a}$$

$\left\{ \begin{array}{l} \text{pokud } a = c \text{ tak zvolíme } \theta = \frac{\pi}{4} \\ \text{pokud } a \neq c \text{ } \frac{2b}{c-a} \in \mathbb{R}, H(\tan 2\theta) = \mathbb{R}. \end{array} \right.$